Finite-Sample Reconstruction:
Rates, Witnesses, and the Honest Bridge

Patrick S. Mahon

1 The Bias Bound and the Algebra Side

1.1 The bias bound

Let $(X, \mathcal{B}, \mu)$ be a probability space and $\mathfrak{m} \subseteq \mathcal{B}$ a sub-σ-algebra. Define the *σ-algebra approximation error*

$$\delta(\mathfrak{m}) \;:=\; \sup_{S \in \mathcal{B}} \inf_{E \in \mathfrak{m}} \mu(S \triangle E). \tag{1}$$

This measures how well $\mathfrak{m}$ approximates $\mathcal{B}$: it is zero if and only if $\mathfrak{m} = \mathcal{B}$ modulo μ-null sets.

Lemma 1.1 (Bias bound)**.** *For every $f \in L^\infty(\mu)$,*

$$\big\|f - \mathbb{E}[f \mid \mathfrak{m}]\big\|_{L^2(\mu)} \;\le\; 2\|f\|_{L^\infty} \cdot \delta(\mathfrak{m})^{1/2}.$$

Proof. Step 1 (Indicators). Fix $S \in \mathcal{B}$. By definition of $\delta(\mathfrak{m})$, there exists $E \in \mathfrak{m}$ with $\mu(S \triangle E) \le \delta(\mathfrak{m})$. Since $\mathbb{1}_E \in L^2(X, \mathfrak{m}, \mu)$ and $\mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]$ is the L^2-best $\mathfrak{m}$-measurable approximation to $\mathbb{1}_S$,

$$\|\mathbb{1}_S - \mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]\|_{L^2}^2 \;\le\; \|\mathbb{1}_S - \mathbb{1}_E\|_{L^2}^2 \;=\; \mu(S \triangle E) \;\le\; \delta(\mathfrak{m}).$$

Step 2 (Bounded f via layer cake). For $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \le M$, the layer cake representation gives $f = \int_{-M}^{M} \mathbb{1}_{S_t}\, dt$ where $S_t = \{x : f(x) > t\}$. Since $\mathbb{E}[\,\cdot \mid \mathfrak{m}]$ is linear and continuous in L^2,

$$f - \mathbb{E}[f \mid \mathfrak{m}] \;=\; \int_{-M}^{M} \big(\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]\big)\, dt.$$

Applying Minkowski's inequality for integrals and Step 1:

$$\big\|f - \mathbb{E}[f \mid \mathfrak{m}]\big\|_{L^2} \;\le\; \int_{-M}^{M} \big\|\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]\big\|_{L^2}\, dt \;\le\; 2M \cdot \delta(\mathfrak{m})^{1/2}. \qquad \square$$

Remark 1.2. The constant 2 is not optimal but the exponent $\frac{1}{2}$ is sharp: taking $X = [0,1]$, $\mathfrak{m} = \{\emptyset, X\}$, $f = \mathbb{1}_{[0,1/2]}$ gives $\delta(\mathfrak{m}) = \frac{1}{2}$ and $\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2} = \frac{1}{2} = \delta(\mathfrak{m})^{1/2}/\sqrt{2}$.

1.2 The three-term decomposition

We now specialise to the delay setting. Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and $h \in L^\infty(\mu)$. Write $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ for the observable σ-algebra at lag L, where $\Phi_h^{(L)}(x) = (h(x), h(Tx), \ldots, h(T^L x))$, and $\mathcal{A}_h^{(L)}$ for the class of degree-D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$. Let $\hat{f}_n^{(L)}$ denote the empirical risk minimiser over $\mathcal{A}_h^{(L)}$.

Lemma 1.3 (Three-term decomposition)**.** *For any $f \in L^\infty(\mu)$, any $L \ge 0$, and any $D \ge 1$:*

$$\big\|f - \hat{f}_n^{(L)}\big\|_{L^2(\mu)} \;\le\; \underbrace{2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}}_{bias} + \underbrace{\inf_{g \in \mathcal{A}_h^{(L)}} \big\|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\big\|_{L^2(\mu)}}_{approx.} + \underbrace{\big\|g_D^* - \hat{f}_n^{(L)}\big\|_{L^2(\mu)}}_{statistical}, \tag{2}$$

where $\delta(L) := \delta(\mathcal{O}_h^{(L)})$ as in (1) and $g_D^* = \operatorname{argmin}_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}$.

Proof. Triangle inequality applied to the decomposition $f - \hat{f}_n^{(L)} = (f - \mathbb{E}[f \mid \mathcal{O}_h^{(L)}]) + (\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g_D^*) + (g_D^* - \hat{f}_n^{(L)})$, with the first term bounded by Lemma 1.1. $\square$

Remark 1.4 (Scope)**.** Lemma 1.1 holds for any sub-σ-algebra $\mathfrak{m}$; no dynamical structure is needed. The dynamical hypothesis enters only through the approximation and statistical terms, which require the image $\Phi_h^{(L)}(X)$ to be a manageable geometric object.

1.3 The observable reverse lower bound

The three-term bound controls the forward error. For the stopping rule to be honest, we also need a reverse bound: when $\delta(L)$ is large, the estimator $\hat{f}_n^{(L)}$ cannot approximate f well, regardless of n.

Lemma 1.5 (Observable reverse lower bound)**.** *There exists a universal constant $c > 0$ such that for any $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \geq \frac{1}{2}$ and any $L \geq 0$:*

$$\left\|\mathbb{1}_S - \hat{f}_n^{(L)}\right\|_{L^2(\mu_n)} \;\geq\; \frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$$

with probability tending to 1 as $n \to \infty$, where $S \in \mathcal{B}$ is the set achieving the supremum in (1) at lag L and $\mu_n = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$ is the empirical measure.

Proof. Step A (Population CE gap). Let $S^* \in \mathcal{B}$ achieve the supremum in $\delta(L) = \sup_S \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E)$. For any $E \in \mathcal{O}_h^{(L)}$:

$$\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)}^2 \;\leq\; \|\mathbb{1}_{S^*} - \mathbb{1}_E\|_{L^2(\mu)}^2 \;=\; \mu(S^* \triangle E) \;\geq\; \delta(L) - \varepsilon$$

for any $\varepsilon > 0$. Taking $\varepsilon \to 0$: $\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)} \geq \delta(L)^{1/2}$.

Since the conditional expectation is the L^2-projection, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} \geq \delta(L)^{1/2}$ for all $g \in \mathcal{A}_h^{(L)} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu)$.

Step B (Empirical vs population norm). By Hoeffding's inequality and the bounded difference property of μ_n, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} - O(n^{-1/2})$ with high probability, uniformly over $g \in \mathcal{A}_h^{(L)}$.

Step C (Empirical norm vs $\hat{f}_n^{(L)}$). Since $\hat{f}_n^{(L)}$ minimises empirical L^2 error over $\mathcal{A}_h^{(L)}$, and $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \delta(L)^{1/2} - O(n^{-1/2})$ for all $g \in \mathcal{A}_h^{(L)}$, we have $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C'n^{-s/(2s+d)}$ with high probability, absorbing the $O(n^{-1/2})$ into the statistical rate. $\square$

Remark 1.6 (Closed loop)**.** Lemmas 1.3 and 1.5 together bound $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)}$ from both sides. The upper bound (Lemma 1.3) says the error is at most $2\|f\|_{L^\infty} \cdot \delta(L)^{1/2} + C \cdot n^{-s/(2s+d)}$. The lower bound (Lemma 1.5) says it is at least $\frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$. The detectability threshold is the n at which the statistical floor $C'n^{-s/(2s+d)}$ drops below $\frac{\delta(L)^{1/2}}{2}$: reconstruction becomes detectable at sample size $n \gtrsim \delta(L)^{-(2s+d)/s}$.

2 Concentration of the Empirical Witness

This section proves that $\hat{\delta}(L, n)$ concentrates around $\delta(L)$. The argument has four steps: uniform boundedness of the function class (Lemma 2.1), McDiarmid concentration around the mean (Proposition 2.3), a Glivenko–Cantelli bound for the LLN gap (Lemma 2.5), and the estimator bias via the Nadaraya–Watson kernel regression (Proposition 2.7). The full convergence is then Theorem 2.9.

Throughout this section the hypotheses of Proposition **??** hold: (T, h) satisfies (G1)–(G2), X is a compact smooth d-manifold, μ has smooth positive density, and $d > 2s$.

2.1 Uniform boundedness

Let $\mathcal{P}_D^{(L)}$ denote the class of degree-D polynomials in the delay coordinates $\Phi_h^{(L)}(x) = (h(x), \ldots, h(T^L x))$, normalised so $\|p\|_{L^2(\mu)} \le 1$. Since $h \in L^\infty$ with $\|h\|_{L^\infty} =: a < \infty$, the delay coordinates lie in the compact box $[-a, a]^{L+1}$.

Lemma 2.1 (Uniform boundedness)**.** *Define*

$$M_{D,L} \;:=\; \binom{D+L+1}{L+1}^{1/2} \cdot (2D+1)^{1/2} \cdot (2a)^{-(L+1)/2} \cdot \rho_{\min}^{-1/2},$$

where $\rho_{\min} = \operatorname{essinf}_X \rho > 0$ *is the infimum of the density of* μ*. Then for every* $p \in \mathcal{P}_D^{(L)}$*:*

$$\|p\|_{L^\infty} \;\le\; M_{D,L} \cdot \|p\|_{L^2(\mu)}.$$

Proof. Rescale to the unit box by setting $y_i = x_i/a$; polynomials of degree $\le D$ in $x \in [-a, a]^{L+1}$ correspond to polynomials $\tilde{p}$ of degree $\le D$ in $y \in [-1, 1]^{L+1}$. Expand $\tilde{p}$ in the tensor-product Legendre basis $\{P_\alpha\}_{|\alpha| \le D}$, where $P_\alpha(y) = \prod_{i=1}^{L+1} P_{\alpha_i}(y_i)$ and $\|P_k\|_{L^\infty[-1,1]} = 1$. Since $\|P_\alpha\|_{L^\infty} = 1$, Cauchy–Schwarz gives

$$|\tilde{p}(y)| \;\le\; \sum_{|\alpha| \le D} |c_\alpha| \;\le\; N(D, L+1)^{1/2} \Big(\sum_{|\alpha| \le D} |c_\alpha|^2 \Big)^{1/2},$$

where $N(D, L{+}1) = \binom{D+L+1}{L+1}$ counts the basis elements. From the L^2 orthogonality $\|P_\alpha\|^2_{L^2[-1,1]} = \prod_i (2\alpha_i + 1)^{-1}$, the coefficient norm satisfies

$$\sum_{|\alpha| \le D} |c_\alpha|^2 \;\le\; (2D+1)\, \|\tilde{p}\|^2_{L^2},$$

since the maximum of $\prod_i (2\alpha_i + 1)$ over $|\alpha| \le D$ is $2D + 1$, achieved at $\alpha = (D, 0, \ldots, 0)$ — the constraint $|\alpha| \le D$ forces all weight onto one coordinate, giving an L-independent maximum. Converting back via $\|\tilde{p}\|_{L^2[-1,1]^{L+1}} = (2a)^{-(L+1)/2} \|p\|_{L^2(\mathrm{vol})}$ and $\|p\|_{L^2(\mathrm{vol})} \le \rho_{\min}^{-1/2} \|p\|_{L^2(\mu)}$ yields the result. □

Remark 2.2. The L-dependence of $M_{D,L}$ is entirely in $N(D, L{+}1)^{1/2} \sim L^{D/2}/\sqrt{D!}$ for $L \gg D$. The factor $(2D+1)^{1/2}$ depends only on degree. For fixed $D = \lceil s \rceil$ and $\|h\|_{L^\infty} < 1$ (so $2a < 1$), $M_{D,L}$ decays exponentially in L; for $\|h\|_{L^\infty} \ge \frac{1}{2}$ (equivalently $2a \ge 1$), $M_{D,L}$ grows polynomially in L. The subsequent arguments require $\|h\|_{L^\infty} \ge \frac{1}{2}$.

2.2 Concentration around the mean

Define the function class

$$\mathcal{F}_{L,D} \;=\; \Big\{ x \mapsto \big(p(x) - \hat{\mathbb{E}}[p \mid \Phi_h^{(L)}](x)\big)^2 \;:\; p \in \mathcal{P}_D^{(L)},\; \|p\|_{L^2(\mu)} \le 1 \Big\},$$

where $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson estimator (Definition 2.6). Then $\hat{\delta}(L, n) = \sup_{f \in \mathcal{F}_{L,D}} \frac{1}{n} \sum_{i=1}^n f(x_i)$.

Proposition 2.3 (Concentration, $\star$)**.** *Assume $d > 2s$. There exist constants $c, C > 0$ depending only on $(d, s, \|h\|_{L^\infty})$ such that for all $t > 0$ and all $n \ge 1$:*

$$P\big(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > t\big) \;\le\; 2 \exp\Big(-\frac{c\, n\, t^2}{M_{D,L}^4 \cdot D^d} \Big). \qquad (\star)$$

In particular, setting $t = \varepsilon_n/2$ with $\varepsilon_n = C'\, n^{-2s/(2s+d)}$:

$$P\big(|\hat{\delta}(L, n) - \mathbb{E}[\hat{\delta}(L, n)]| > \varepsilon_n/2\big) \;\le\; 2 \exp\Big(-c'\, n^{(d-2s)/(2s+d)} \Big) \;\to\; 0.$$

Proof. Step 1 (Boundedness). By Lemma 2.1, every $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \leq 1$ satisfies $\|p\|_{L^\infty} \leq M_{D,L}$. Hence every $f = (p - \hat{\mathbb{E}}[p|\Phi_h^{(L)}])^2 \in \mathcal{F}_{L,D}$ satisfies $\|f\|_{L^\infty} \leq 4M_{D,L}^2 =: B_{L,D}^2$.

Step 2 (McDiarmid). Replacing one observation x_i with x_i' changes $\hat{\delta}(L,n)$ by at most $B_{L,D}^2/n$. By the McDiarmid bounded-difference inequality:

$$P\big(|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]| > t\big) \;\leq\; 2\exp\Big(-\frac{2n^2t^2}{n \cdot (B_{L,D}^2/n)^2}\Big) \;=\; 2\exp\Big(-\frac{n\,t^2}{2B_{L,D}^4/n}\Big).$$

Step 3 (Rademacher bias). By the symmetrisation inequality [?]:

$$\mathbb{E}[\hat{\delta}(L,n)] - \delta(L) \;\leq\; 2\,\mathfrak{R}_n(\mathcal{F}_{L,D}).$$

By the Ledoux–Talagrand contraction lemma, since $f = (p-\cdot)^2$ is $2B_{L,D}$-Lipschitz in the second argument:

$$\mathfrak{R}_n(\mathcal{F}_{L,D}) \;\leq\; 2B_{L,D}\,\mathfrak{R}_n(\mathcal{P}_D^{(L)}).$$

In the Takens regime the image $\Phi_h^{(L)}(X)$ is a d-dimensional submanifold, so the VC dimension of $\mathcal{P}_D^{(L)}$ restricted to $\Phi_h^{(L)}(X)$ is $O(D^d)$ [?]. Hence $\mathfrak{R}_n(\mathcal{P}_D^{(L)}) \leq C\sqrt{D^d/n}$, giving $\mathfrak{R}_n(\mathcal{F}_{L,D}) \leq C\,M_{D,L}\sqrt{D^d/n}$.

Step 4 (Combined). Combining Steps 2–3 and absorbing constants:

$$P\big(|\hat{\delta}(L,n) - \delta(L)| > t\big) \;\leq\; 2\exp\Big(-\frac{c\,n\,t^2}{M_{D,L}^4\,D^d}\Big).$$

Setting $t = \varepsilon_n/2 = C'n^{-2s/(2s+d)}/2$, the exponent becomes

$$c\,n\,\varepsilon_n^2/(M_{D,L}^4\,D^d) \;\propto\; n^{1-4s/(2s+d)} \;=\; n^{(d-2s)/(2s+d)}.$$

This tends to $+\infty$ if and only if $d > 2s$, proving the final claim. $\square$

Remark 2.4 (The $d > 2s$ hypothesis)**.** The condition $d > 2s$ is necessary for the concentration to be useful at the rate ε_n. It is not an artefact: when $d \leq 2s$, the function class $\mathcal{F}_{L,D}$ is too rich for n samples to resolve the fine structure of $\delta(L)$ at the required scale. The fix via localised Rademacher complexity [?] is technically available but left to future work; Paper IV takes $d > 2s$ as a standing hypothesis.

2.3 Source 1: the LLN gap

Lemma 2.5 (LLN gap)**.** *With notation as above:*

$$\sup_{p \in \mathcal{P}_D^{(L)}} \Big| \|p - \mathbb{E}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu_n)}^2 - \|p - \mathbb{E}[p \mid \Phi_h^{(L)}]\|_{L^2(\mu)}^2 \Big| \;\leq\; C\sqrt{D^d/n}$$

almost surely as $n \to \infty$, where C depends on $(d, D, M_{D,L})$.

Proof. The class $\{(p - \mathbb{E}[p|\Phi_h^{(L)}])^2 : p \in \mathcal{P}_D^{(L)}\}$ is a VC class of dimension $O(D^d)$. The uniform Glivenko–Cantelli theorem [?] gives a.s. convergence of the empirical mean to the population mean, at rate $O(\sqrt{D^d/n})$. $\square$

2.4 Source 2: estimator bias

The estimator $\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}]$ is the Nadaraya–Watson (NW) kernel regression estimator.

Definition 2.6 (Nadaraya–Watson estimator)**.** For bandwidth $h_n > 0$ and kernel $K : \mathbb{R}^{L+1} \to \mathbb{R}_{\geq 0}$ (compactly supported, symmetric, $\int K = 1$):

$$\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}(x)] \;:=\; \frac{\sum_{j=1}^n K\big(\frac{\Phi_h^{(L)}(x)-\Phi_h^{(L)}(x_j)}{h_n}\big)\, p(x_j)}{\sum_{j=1}^n K\big(\frac{\Phi_h^{(L)}(x)-\Phi_h^{(L)}(x_j)}{h_n}\big)}.$$

Proposition 2.7 (Source 2 estimator bias)**.** *Assume:*

(i) $T, h \in C^r$ with $r \geq 2$, μ smooth positive density ρ with $\rho_{\min} > 0$.

(ii) $\Phi_h^{(L)}$ has full rank μ-a.e. (follows from (G1)–(G2) and Lemma **??***).*

(iii) T is exponentially mixing with rate $\lambda > 0$ and $\|h\|_{L^\infty} \geq \frac{1}{2}$.

(iv) The NW bandwidth satisfies $h_n \sim n^{-1/(2\beta+d_{\mathrm{eff}}(L))}$ with $d_{\mathrm{eff}}(L) = \min(L+1, d)$ and $\beta \geq 1$.

Then there exists $C_2(D, L) > 0$ such that for all $p \in \mathcal{P}_D^{(L)}$:

$$\mathbb{E}\|\hat{\mathbb{E}}[p \mid \Phi_h^{(L)}] - \mathbb{E}[p \mid \Phi_h^{(L)}]\|^2_{L^2(\mu)} \;\leq\; C_2(D, L) \cdot n^{-2\beta/(2\beta+d_{\mathrm{eff}}(L))},$$

uniformly in $p \in \mathcal{P}_D^{(L)}$, where $C_2(D, L) = 2M_{D,L}^2 C_1^2 A^{2\beta}$ and A is a constant depending on $(\beta, d_{\mathrm{eff}}(L), \rho_{\min})$.

Proof. The proof is a four-link chain.

Link 1 (Smooth disintegration). Under conditions (i)–(ii), the Rokhlin disintegration $\mu = \int \mu_z \, d((\Phi_h^{(L)})_*\mu)(z)$ has fibre measures μ_z supported on the smooth fibres $F_z = \{\Phi_h^{(L)} = z\}$. An implicit-function-theorem argument (three sub-steps: transfer to common surface via a C^{r-1} diffeomorphism; bound weight variation; bound Jacobian factor) gives:

$$\|\mu_z - \mu_{z'}\|_{\mathrm{TV}} \;\leq\; C_1\,|z - z'|^\beta, \qquad \beta = 1 \text{ (Lipschitz floor)},$$

where C_1 depends on $(\|T\|_{C^r}, \|h\|_{C^r}, \rho_{\min})$. (The optimal exponent $\beta = r - 1$ follows from the full Taylor expansion of the fibre weight; the floor $\beta = 1$ is sufficient for convergence.)

Link 2 (Bernstein–Markov). From Link 1 and Lemma 2.1: for $p \in \mathcal{P}_D^{(L)}$ with $\|p\|_{L^2(\mu)} \leq 1$,

$$\big|\mathbb{E}[p \mid \Phi_h^{(L)} = z] - \mathbb{E}[p \mid \Phi_h^{(L)} = z']\big| \;=\; \Big|\int p\, d\mu_z - \int p\, d\mu_{z'}\Big| \;\leq\; M_{D,L} \cdot C_1 \cdot |z - z'|^\beta.$$

Hence $z \mapsto \mathbb{E}[p \mid \Phi_h^{(L)} = z]$ is Hölder(β) with constant $\Lambda_{D,L} := M_{D,L} C_1$, uniformly over $p \in \mathcal{P}_D^{(L)}$.

Link 3 (NW bias-variance balance). Standard NW theory [**?**] for a Hölder(β) regression function on a $d_{\mathrm{eff}}(L)$-dimensional domain gives:

$$\mathbb{E}\|\hat{\mathbb{E}}[p|\Phi_h^{(L)}] - \mathbb{E}[p|\Phi_h^{(L)}]\|^2_{L^2(\mu)} \;\leq\; \Lambda_{D,L}^2\, h_n^{2\beta} + \frac{C_K\, M_{D,L}^2}{n\, h_n^{d_{\mathrm{eff}}(L)}\, f_{\min}},$$

where C_K depends on the kernel K and $f_{\min}$ is the minimum of the marginal density of $(\Phi_h^{(L)})_*\mu$. The $M_{D,L}^2$ factor appears in both terms. Balancing: $\Lambda_{D,L}^2 h_n^{2\beta} = C_K M_{D,L}^2 / (n h_n^{d_{\mathrm{eff}}(L)} f_{\min})$ gives $h_n^* \sim A \cdot n^{-1/(2\beta+d_{\mathrm{eff}}(L))}$ where A depends only on $(\beta, d_{\mathrm{eff}}(L), C_K, f_{\min}, C_1)$ — not on D, L, or p. The $M_{D,L}^2$ factors cancel from both sides of the balance equation.

Link 4 (Uniformity). Substituting h_n^*, both terms contribute $C_2(D, L) \cdot n^{-2\beta/(2\beta+d_{\mathrm{eff}}(L))}$ with $C_2(D, L) = 2M_{D,L}^2 C_1^2 A^{2\beta}$. Since the bound is p-free, it holds uniformly over $\mathcal{P}_D^{(L)}$. □

Remark 2.8 (Scope)**.** Proposition 2.7 is proved under exponential mixing with $\|h\|_{L^\infty} \geq \frac{1}{2}$. The polynomial mixing case and the regime $\|h\|_{L^\infty} < \frac{1}{2}$ require a different estimator and are left open.

2.5 Full convergence

Theorem 2.9 (Full convergence $\hat{\delta}(L,n) \to \delta(L)$)**.** *Under the hypotheses of Proposition 2.7 and with $d > 2s$:*

$$\mathbb{E}\big|\hat{\delta}(L,n) - \delta(L)\big| \;\le\; C\sqrt{D^d/n} \;+\; C_2(D,L)^{1/2} \cdot n^{-\beta/(2\beta+d_{\mathrm{eff}}(L))}$$

and $P(|\hat{\delta}(L,n) - \delta(L)| > \varepsilon_n) \to 0$ exponentially, where $\varepsilon_n = C' n^{-2s/(2s+d)}$.

Proof. Decompose:

$$|\hat{\delta}(L,n) - \delta(L)| \;\le\; \underbrace{|\hat{\delta}(L,n) - \mathbb{E}[\hat{\delta}(L,n)]|}_{\text{concentration, } (\star)} \;+\; \underbrace{|\mathbb{E}[\hat{\delta}(L,n)] - \delta(L)|}_{\text{bias}}$$

The bias splits as Source 1 + Source 2: Source 1 is bounded by $C\sqrt{D^d/n}$ (Lemma 2.5); Source 2 is bounded by $C_2(D,L)^{1/2} \cdot n^{-\beta/(2\beta+d_{\mathrm{eff}}(L))}$ (Proposition 2.7). The concentration term is bounded by $(\star)$ (Proposition 2.3). Combining at $t = \varepsilon_n/2$ gives the probability statement. ☐